

Reflection Paper: Segmenting Wholesale Clients with Clustering

1. What I Implemented

I segmented wholesale distribution clients using their annual spending across six product categories — Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen — as the input features. Before clustering, I scaled all six spending columns so that high-volume categories like Fresh and Grocery would not dominate the distance calculations simply because they involve larger raw numbers.

My primary algorithm was K-Means with $k = 5$, which I chose to match the number of clusters used for the comparison model. I also implemented Agglomerative (hierarchical) Clustering, also with 5 clusters, as my additional algorithm. Both models were trained on the same scaled feature set so their outputs could be compared directly. I evaluated both using Silhouette Score and the Davies-Bouldin Index, and ran a sanity check on the first three customers in the dataset to see whether the two algorithms grouped the same customers together.

2. Segment Interpretation

Based on the K-Means results, three of the clusters stood out as clearly distinguishable customer profiles:

Cluster A — Fresh-dominant buyers: these clients spend heavily on Fresh products relative to the other categories, with comparatively low spending on Detergents_Paper and Grocery. This profile is typical of restaurants, cafes, or other food-service businesses that need a constant supply of perishable ingredients. Business action: the distributor could offer more frequent, smaller-batch delivery schedules and prioritize freshness/cold-chain logistics for this segment rather than bulk discounts.

Cluster B — Grocery and Detergents_Paper-heavy buyers: these clients spend disproportionately on Grocery and Detergents_Paper, with more moderate Fresh spending. This looks like traditional retailers or supermarkets that stock shelf-stable goods and household supplies. Business action: the distributor could set up volume-based bulk pricing tiers and loyalty discounts for this segment, since their purchasing is steady and predictable rather than perishable-driven.

Cluster C — Low overall spenders: this cluster spends relatively little across all six categories compared to the others. These are likely small, low-volume clients such as small corner stores. Business action: the distributor could bundle smaller, mixed-category orders together to reduce per-delivery cost, or use targeted promotions to try to grow these accounts into higher-volume customers.

3. Understanding K-Means

In my own words, K-Means is an unsupervised clustering algorithm that groups data points into a fixed number of clusters, k , based on how close they are to each other in feature space. It works through an iterative assign-and-update loop:

- First, k initial centroids (cluster centers) are chosen, often randomly or with a smarter method like `k-means++`.
- Assign step: every data point is assigned to the nearest centroid, based on distance (typically Euclidean distance), which forms k clusters.
- Update step: each centroid is recalculated as the mean of all the points currently assigned to its cluster, which moves the centroid to the center of its group.
- These two steps repeat — reassigning points to the nearest (now updated) centroid, then recalculating centroids — until the assignments stop changing or the centroids move by less than some small threshold.

The end result is k clusters where points within a cluster are close to their shared centroid, and, ideally, clearly separated from points in other clusters. Because the algorithm depends on the initial centroid placement, it's common to run K-Means multiple times with different starting points and keep the best result.

4. My Second Algorithm

Choice: I chose Agglomerative (hierarchical) Clustering as my second algorithm. I chose it because, unlike K-Means, it doesn't rely on randomly initialized centroids or assume clusters are roughly spherical, so it gave me a useful point of comparison for whether the wholesale customer segments were sensitive to the clustering method itself.

From my research, Agglomerative Clustering works bottom-up: it starts by treating every single data point as its own cluster, then repeatedly merges the two closest clusters together based on a linkage criterion (such as average, complete, or Ward linkage) until only the desired number of clusters, k , remains. This merging process can also be visualized as a dendrogram, which shows the full hierarchy of merges.

Advantage: it does not require specifying k in advance if you're willing to inspect the dendrogram, and it can capture non-spherical or unevenly sized clusters better than K-Means in some cases, since it doesn't rely on distance-to-centroid alone.

Limitation: it is more computationally expensive than K-Means (its complexity grows quickly with the number of data points), and once two clusters are merged, that merge cannot be undone later in the process, so early mistakes in the hierarchy can propagate.

On this dataset, Agglomerative Clustering's Silhouette Score (0.218) was lower than K-Means' Silhouette Score (0.283), and its Davies-Bouldin Index (1.325) was higher than K-Means' (1.270) — and since a higher Silhouette Score and a lower Davies-Bouldin Index both indicate better-defined clusters, K-Means produced the more clearly separated, more cohesive clusters on this particular dataset. The sanity check on the first three customers reflected some disagreement between the two methods: K-Means grouped all three of the first customers into the same cluster (Cluster 0), but Agglomerative only agreed that customers 1 and 2 belonged together (both Cluster 4) — it split customer 3 off into a different cluster (Cluster 0) instead of grouping it with the first two. (Note that the cluster label numbers themselves aren't comparable across algorithms; what matters is which customers end up grouped together, not the number assigned to the group.) This shows the two methods don't always agree on how individual customers should be segmented, even on the same scaled data.

5. Metrics Comparison

Algorithm	Silhouette Score	Davies-Bouldin Index
K-Means ($k=5$)	0.283	1.270
Agglomerative ($k=5$)	0.218	1.325

6. My Findings

Based on these results, I would recommend K-Means ($k = 5$) for this wholesale segmentation task. Its Silhouette Score of 0.283 was higher than Agglomerative Clustering's 0.218, and its Davies-Bouldin Index of 1.270 was lower than Agglomerative's 1.325 — on both metrics, K-Means produced clusters that were more internally cohesive and more clearly separated from one another. K-Means is also computationally cheaper and easier to re-run as new customer data comes in, which matters for a distributor that would want to periodically re-segment its client base as spending patterns change.

That said, I wouldn't dismiss Agglomerative Clustering entirely — its dendrogram-based structure could still be useful as an exploratory tool to sanity-check how many clusters actually make sense before committing to a k value for K-Means, or to double check that the K-Means segments aren't an artifact of unlucky centroid initialization. But as the primary segmentation method the distributor uses to drive business decisions like delivery scheduling and bulk pricing, K-Means is the stronger and more practical choice given its higher-quality clustering metrics on this dataset.